

Sukhrob Ilyosbekov

Senior Software Engineer · Full-Stack & AI

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SUMMARY

Senior Software Engineer with 9+ years shipping scalable web apps, SaaS platforms, AI products, and multiplayer games across healthcare, real estate, logistics, and gaming. Deep experience with HIPAA-compliant systems, real-time architectures, and LLM-powered applications; published computer-vision research (arXiv) in explainable AI for medical imaging.

TECHNICAL SKILLS

Languages: C#, C++, Python, JavaScript, TypeScript, Kotlin

Backend: ASP.NET Core, EF Core, SignalR, Node.js, Bun, NestJS, FastAPI, ElysiaJS

Frontend & Mobile: React, Next.js, Angular, Blazor, Tailwind CSS, Kotlin Multiplatform, .NET MAUI, React Native

Data: PostgreSQL, MS SQL, MongoDB, Redis, Firebase, DynamoDB, Prisma

AI/ML: PyTorch, TensorFlow, OpenCV, YOLO, Claude/GPT/DeepSeek APIs, MCP, agentic workflows

Game Dev: Unity, Godot, PhaserJS, Colyseus, Metamod:Source 2

Cloud & DevOps: AWS (Amplify, S3, Lambda, Cognito), Azure, Docker, Kubernetes, GitHub Actions

EDUCATION

Northeastern University , M.S. in Computer Science, Boston, MA	<i>Jan 2024 – May 2026</i>
Suffolk University , B.S. in Computer Science, Boston, MA	<i>Sep 2021 – May 2023</i>
INTI International College Subang , B.S. in Computer Science, Malaysia	<i>Sep 2019 – Jul 2021</i>
TUIT , B.S. in Software Engineering, Uzbekistan	<i>Sep 2017 – Jun 2019</i>

PROFESSIONAL EXPERIENCE

Senior Software Engineer | [EmTech Care Labs](#) | Portland, ME *Jan 2025 – Jun 2025*

- Architected and shipped a HIPAA-compliant care-coordination platform on AWS Amplify serving 200+ caregivers and patients, with real-time messaging, push notifications, and shared care-plan editing for collaborative Alzheimer's care.
- Led AWS Amplify Gen 1 to Gen 2 migration across the codebase, cutting deploy time from 12 to 7 minutes (40%) and reducing monthly CI spend by ~\$800.
- Consolidated 3 fragmented frontends into a single TypeScript monorepo and design system of 40+ shared components, cutting new-screen build time from days to hours and eliminating UI drift between web and mobile.
- Hardened the platform for HIPAA: field-level PHI encryption, audit logging via DynamoDB Streams, and Cognito-based RBAC; cleared a third-party HIPAA readiness assessment with zero critical findings.
- Owned end-to-end delivery as the sole senior engineer: scoped requirements with clinicians, shipped 10+ features in 6 months, and mentored 2 junior engineers through onboarding and code review.

Senior Software Engineer | [AllFactors](#) | Santa Clara, CA *July 2022 – Dec 2023*

- Drove migration of a legacy ASP.NET Web Forms real-estate analytics application to Blazor WebAssembly, cutting initial page load from 4.2s to 2.1s, shrinking bundle size by 35%, and removing all ViewState dependencies.
- Designed and shipped a Blazor component library of 30+ responsive UI primitives (data grids, charts, filters), reducing new-page implementation time from ~3 days to ~6 hours and standardizing dashboards across the product.

- Built a real-time market-data pipeline using SignalR and SQL Server change tracking, pushing pricing and inventory updates to 500+ concurrent users with sub-200ms end-to-end latency.
- Introduced an automated test suite (xUnit + Playwright) covering critical user flows, raising code coverage from 12% to 68% and catching 30+ regressions before release.

Software Engineer | [Smart Meal Service](#) | Russia

Sep 2020 – Oct 2021

- Developed a robotic cashier and self-service kiosk in C# / .NET processing 500+ daily transactions, integrating with POS terminals and payment hardware over TCP/IP and serial protocols.
- Shipped a customer mobile ordering app (Xamarin) and admin dashboard (ASP.NET Core + Angular), cutting average kitchen ticket time by ~25% through automated order routing.
- Built a hardware abstraction layer that let the same kiosk software run across 3 machine variants, reducing per-store deployment from ~2 days to under 4 hours.

Game Developer | [Pentalight Technology](#) | Malaysia

Mar 2020 – Feb 2021

- Built multiplayer networking for a VR smart-city simulation in Unity, supporting 20+ concurrent users with sub-100ms state sync via MLAPI and SteamVR.
- Implemented spatial audio, hand-tracking, and HUD systems, plus a spectator/desktop client so non-VR participants could observe and interact with the world in real time.
- Authored an asset pipeline automating FBX import, LOD generation, and lightmap baking, cutting per-scene iteration from 4 hours to 20 minutes.

Earlier: Game Developer at EC Dev Team, Uzbekistan (2016–19) – Led RTS game development in Clausewitz Engine. Freelance Developer at Freelancer.com (2019–20) – .NET and Python NLP applications.

FEATURED PROJECTS

LogisticsX

<https://logisticsx.app> | [GitHub](#)

.NET 10, EF Core, MediatR, SignalR, Angular 21, Kotlin KMP, PostgreSQL, Claude API, Stripe, Docker

- Built an agentic dispatcher on the Claude API (custom MCP server, tool-use loop, structured reasoning traces) that autonomously matches loads to trucks, verifies HOS/FMCSA compliance, and optimizes multi-stop routes – replacing ~15-minute manual dispatch decisions with sub-30-second AI ones.
- Shipped 4 Angular 21 portals (admin, dispatcher, customer, broker) plus a Kotlin Multiplatform driver app (Android/iOS) on a multi-tenant .NET 10 / EF Core / SignalR backend, with DAT, Truckstop, 123Loadboard load-board integrations, Samsara/Motive ELD, and Stripe Connect payouts.

Meat.gg

<https://meat.gg>

ElysiaJS, Next.js, PostgreSQL, Prisma, Bun, C++23, Metamod:Source 2, Docker, Stripe

- Launched a community platform for CS2 servers reaching 30K+ registered users and 1K+ daily active players, with profiles, messaging, friend graph, and a Stripe-backed cosmetic shop processing 100+ monthly transactions.
- Wrote a Metamod:Source 2 server plugin in C++23 that bridges CS2 game servers with the ElysiaJS backend over WebSockets, enabling in-game ban/report/admin workflows with sub-500ms round-trip without relying on RCON.

OGStack

<https://ogstack.dev> | [GitHub](#)

Next.js 16, Elysia, Bun, PostgreSQL, Prisma, Satori, Cloudflare R2, Stripe, Docker

- SaaS that auto-generates on-brand Open Graph images for any URL via a single meta tag, serving cached images under 200ms p95 via Cloudflare R2 + CDN; generated 10K+ images during beta.
- Built a multi-model AI routing layer (Claude, GPT, DeepSeek) for page-content analysis and brand extraction, plus Flux 2 background generation and a Satori-based JSX→SVG→PNG renderer with Stripe-billed subscriptions.

DepVault

<https://depvault.com> | [GitHub](#)

Next.js, ElysiaJS, PostgreSQL, Prisma, Bun, .NET AOT CLI, Docker

- Dependency scanner and encrypted secrets vault covering 8+ language ecosystems (npm, PyPI, NuGet, Cargo, Maven, Go, RubyGems, Composer) with CVE detection via OSV.dev, AES-256-GCM envelope encryption, one-time secret sharing, and a .NET AOT CLI for CI/CD token injection.

- Designed a streaming SBOM ingestion pipeline (Elysia + Bun) that scans 1K+ dependencies per project in under 5 seconds, with deduplicated vulnerability normalization across OSV, GHSA, and NVD feeds.

Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

FastAPI, Next.js, PyTorch, OpenCV, OHIF Viewer, PostgreSQL

- HIPAA-compliant platform that connects to hospital PACS over DICOM C-FIND/C-MOVE to retrieve X-ray, CT, and MRI studies, running PyTorch detection models (pneumonia on CXR, intracranial hemorrhage on head CT) with overlays rendered in the OHIF Viewer.
- Implemented end-to-end PHI handling – on-the-fly de-identification, audit logging, role-based study access – and achieved sub-2s round-trip from PACS query to AI overlay on a 512-slice CT.

Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

YOLO, Moondream2 LLM, FastAPI, Angular

- Built an end-to-end book-detection pipeline: YOLO segmentation isolates individual book spines from shelf photos at ~92% mAP@0.5, then Moondream2 VLM extracts title and author with ~85% string-match accuracy.
- Wrapped the pipeline in a FastAPI service with an Angular frontend supporting upload, manual edit, and bulk export to Goodreads/LibraryThing CSV.

MelanomaNet

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, GradCAM++, OpenCV

- Trained an EfficientNet V2 classifier on the ISIC 2019 dataset (25K+ dermoscopic images, 9 diagnostic categories), achieving 85.61% accuracy and a 0.8564 weighted F1 score.
- Layered GradCAM++ heatmaps with a novel ABCDE-criterion (Asymmetry, Border, Color, Diameter, Evolution) decomposition for clinician-readable interpretability; method published on arXiv (2512.09289).